SALES DATA PROJECT

Analysing customer sales behaviour and measuring segment contribution to revenue and profit

Prepared: March 30, 2026 | **Status:** IN PROGRESS | **Phase:** Data Analysis & Visualization

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PROJECT OVERVIEW

The Sales Data project analyses customer sales behaviour and measures contribution of each customer or product segment to overall revenue and profit.

The project follows a structured pipeline from raw data ingestion through cleaning and transformation to interactive visual reporting.

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PROJECT OBJECTIVES

IDENTIFY PATTERNS

Detect patterns in customer sales behaviour across time periods

MEASURE CONTRIBUTION

Calculate individual and segment-level contribution to total sales and profit

COMPUTE METRICS

Calculate profit, profit percentage, and average sales values

DELIVER DASHBOARD

Build interactive Power BI dashboard for business stakeholders

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WORK COMPLETED

PHASE 1: DATA LOADING

Loaded raw Excel file into Python environment using VS Code and pd.read_excel() for processing

PHASE 2: DATA CLEANING

Detected 15 null values in Order Date column using df.isnull().sum(). Fixed date parsing with errors='coerce' and removed blank rows with df.dropna()

PHASE 3: FEATURE ENGINEERING

Derived new columns: Profit = Sales – Cost, Year and Month extracted from Order Date. Exported cleaned dataset as CSV

PHASE 4: SQL INTEGRATION & DASHBOARD

Connected cleaned CSV to SQL database. Built interactive Power BI dashboard with key metrics

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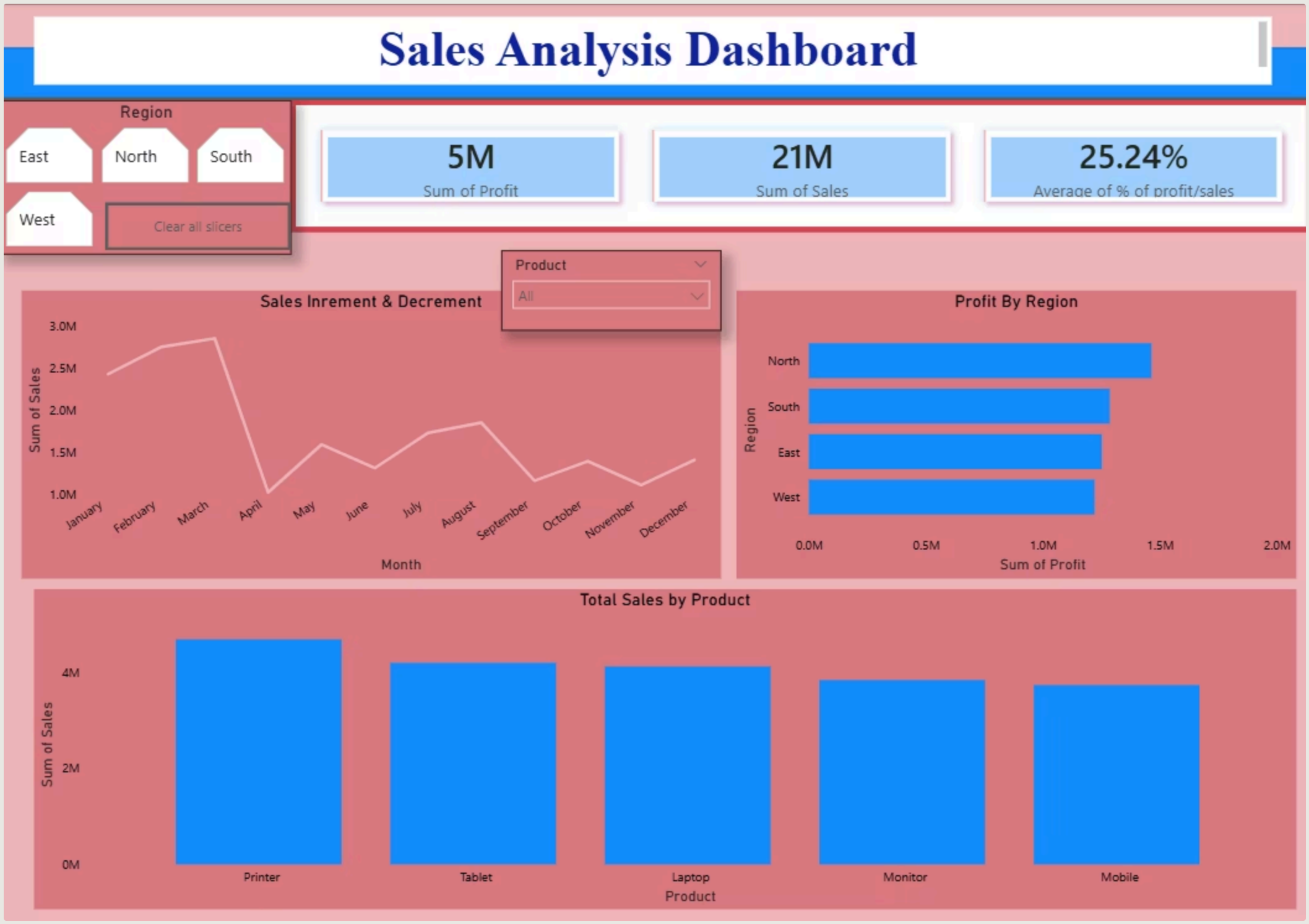
DASHBOARD METRICS DELIVERED



TOTAL SALES

Sum of all sales transactions across the dataset

AVERAGE SALES

Mean sales value per transaction or customer

SALES BY PRODUCT

Breakdown of revenue contribution by product category

PROFIT

Calculated as Sales minus Cost for each record

PROFIT % OF SALES

Profit expressed as a percentage of total sales revenue

AVG. SALES PERCENTAGE

Average percentage share of sales across segments

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TOOLS & TECHNOLOGIES

PYTHON (PANDAS)

Data loading, cleaning, and feature engineering

VS CODE

Development environment for writing and running Python scripts

SQL

Data storage and querying of the cleaned dataset

POWER BI

Interactive dashboard creation and data visualization

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NEXT STEPS

FINALIZE DASHBOARD

Validate all Power BI visuals with stakeholders

ADD SEGMENTATION

Include customer segmentation by region or category

PUBLISH & AUTOMATE

Publish dashboard for team access and schedule automated data refresh

DOCUMENT PIPELINE

Document end-to-end process for reproducibility and future updates

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PROJECT STATUS SUMMARY

PROGRESS OVERVIEW

The project is in Phase 4 of 4, with data loading, cleaning, and feature engineering complete. The Power BI dashboard is under development.

KEY ACHIEVEMENTS

- 15 null values detected and resolved in Order Date column
- 3 new calculated fields added: Profit, Year, and Month
- Interactive dashboard displaying 6 key metrics built

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KEY METRICS OVERVIEW

15

NULL VALUES FIXED

Empty values in Order Date column detected and resolved

3

NEW FIELDS CREATED

Profit, Year, and Month derived from raw data

6

DASHBOARD METRICS

Key financial indicators displayed in Power BI

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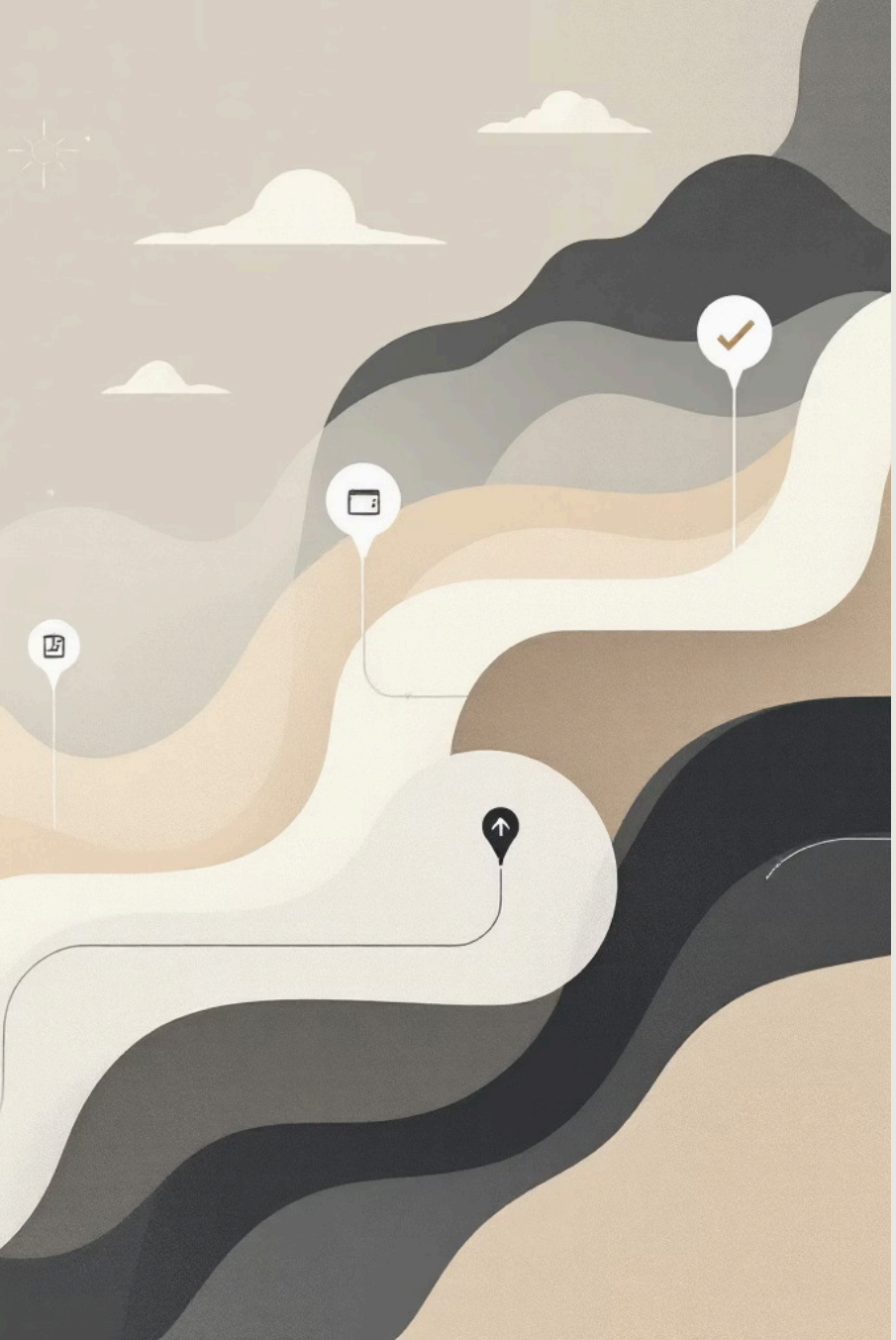

PROJECT STATUS

3 OF 4 COMPLETE

Data loading, cleaning, and feature engineering phases finished

IN PROGRESS

SQL integration and Power BI dashboard development ongoing

NEXT

Finalize visuals, add segmentation, publish dashboard, document pipeline

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